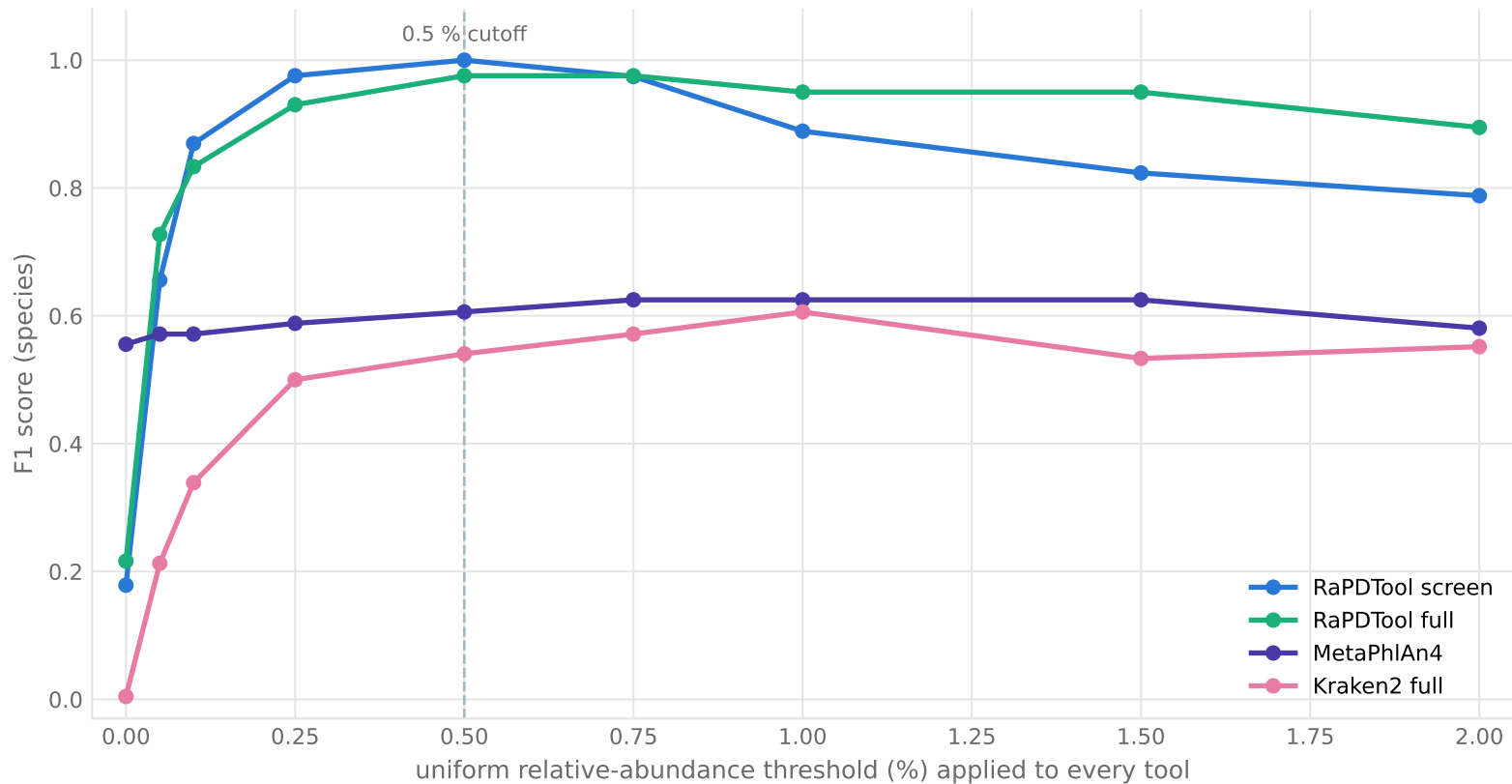


**Even the FOCUS abundance profile overtakes MetaPhlAn on F1 under a shared cutoff
(detection F1 is already 1.0 from the mash table)**



mock_In_30M, species rank. MetaPhlAn wins only at 0 % (its output is already marker-filtered); at any shared threshold ≥ 0.1 % RaPDTool leads, reaching F1 = 1.0 at 0.5 %. Above 1 % recall falls: two genomes are <1 % abundance by design.